Reliable Strictly Local Floorplan Editing under Imperfect Controls

Anonymous CVPR submission

Abstract

Floorplan editors often report edit accuracy without establishing that unrelated architecture remains unchanged. We study a complementary requirement: reliable local execution under an explicit control interface. A request consists of an old CAD state, a declared edit region, and a typed control change. The output may change only inside the declared region and must remain structurally legal. We evaluate this contract in two regimes. Under target-derived oracle controls, FloorFlow reaches Delta-IoU 0.956703–0.999801 and Delta-MSE 0.000019–0.000096 across five operations while preserving outside pixel, room, polygon, wall, opening, and unaffected-edge content exactly. Under target-free predicted controls, it preserves the same outside contract, produces structurally plausible plans over five repeated validation seeds, and abstains from every unsupported or ambiguous request in a development-exposed prompt audit. We keep these safety properties separate from reference fidelity: predicted Room macro IoU is 0.269701, raw watertightness is 0.816701, and an older-model semantic diagnostic ranges from 0.00 to 0.92. The contribution is a scoped, auditable editing result and a reproducible boundary between locality, legality, and semantic realization, not a claim of complete benchmark superiority.

1. Introduction

A floorplan edit is not merely a plausible new drawing. It is a controlled transition from an existing plan: the requested operation must be realized, the unaffected architecture must remain untouched, and the result must remain legal. These requirements are usually conflated in end-to-end metrics. A global regeneration can score well on an image or vector comparison while silently moving walls outside the requested region. Conversely, a perfectly local change can preserve geometry while failing to perform the requested semantic operation.

We treat locality, topology legality, edit support, and semantic realization as separate measurement axes. The input is an old plan, its control state, a declared edit region, and a typed operation. FloorFlow composes a control change only inside the declared domain, restores the outside layers exactly, and audits the resulting topology. Unsupported wall movement, ambiguous global scope, and conflicting instructions are routed to abstention rather than forced through the generator.

The evaluation deliberately separates two evidence regimes. Oracle controls use target-derived after-edit controls and establish an editing upper bound; they are not deployable. Predicted controls are target-free at inference and expose the real operating point, including weak reference fidelity and mixed semantic success. A matched reconstruction wrapper, an outside-frozen editing-aware control, deterministic composition-policy arms, and protocol attacks test whether the observed locality and edit support can be explained by freezing alone.

Our contributions are:

- a strictly local typed-edit protocol with an explicit outside domain and zero-tolerance preservation audit;
- a scoped oracle-control anchor over 2,479 edits, five operations, and five repeated execution seeds;
- a target-free reliability audit that separates execution safety, structural plausibility, raw topology, and reference fidelity;
- matched controls and a deterministic composition ablation showing where the measured gain comes from and where it does not;
- a reproducible evidence release whose numerical claims are bound to a source registry, frozen file identities, and case-level figure provenance.

The paper does not claim complete HouseMind or RPLAN superiority, full-CAD SOTA, deployable free-text editing, or fresh held-out generalization. Those boundaries are part of the result.

2. Related Work

HouseMind unifies floorplan understanding, generation, and editing with discrete room-instance representations and reports edit and graph metrics on an RPLAN natural-language protocol [4]. We use its published Delta-IoU and Delta-MSE values only as cross-dataset context. The public table does not expose strict outside preservation, locality violations, negative controls, or attack sensitivity, so it cannot certify the contract studied here.

Raster2Seq reconstructs vector plans from raster inputs [5]. We use an outside-frozen raster-to-vector redraw as a matched reconstruction wrapper. This is a strong diagnostic against the concern that a locality wrapper alone produces the reported result, but it is not a native editing model and we do not attribute editing awareness to it. CE2EPlan studies boundary-conditioned vector generation [3]; generation evidence is not interpreted as editing evidence.

Floorplan recognition and vectorization provide the geometric substrate. CubiCasa5K supplies heterogeneous annotated plans [1], while graph representations such as GSDiff inform structured evaluation [2]. We do not pool aligned, canonical-instance, and typed graph protocols, and we report graph values as diagnostics rather than a ranking.

3. Problem and Protocol

3.1. Control Transition

Let x_0 be the old visible plan and c_0 its control state. A request contains a normalized or control-grid edit region R and a typed operation a. The edited control is $c_1 = T_a(c_0)$, and the model outputs

$$x_1 = f(x_0, c_1, R). \tag{1}$$

The old control state is an allowed input because engineering edits are applied relative to an existing drawing and its constraints. The contract therefore asks two separate questions: is the request supported, and does the output realize the requested change without changing the rest of the plan?

3.2. Strict Outside Domain

The valid outside domain is

$$O = P \wedge \neg R \wedge \neg D_1(W_0 \vee W_1), \tag{2}$$

where P is the footprint, W_0, W_1 are pre- and post-edit wall support, and D_1 is one-pixel dilation. Pixel, room-label, room-mask, polygon, wall, opening, and unaffected-edge content outside O is compared exactly. A change there is a contract failure, not a small quality loss. The changed-control mask is an anchor for the requested edit and cannot expand the allowed domain.

3.3. Operations and Abstention

The supported operations are room-type change, room addition, room removal, opening deletion, and furniture deletion. Unsupported wall movement, ambiguous global scope, and conflicting instructions trigger safe abstention. Abstention returns the pre-edit plan and is never counted as a semantic edit.

4. FloorFlow

4.1. Typed Control Routing

FloorFlow represents rooms, walls, openings, furniture, and support regions as typed control families. A typed router selects the family and spatial support affected by the request. The routing policy used in the main evidence is deterministic and adds no learned parameters over the matched no-graph arm. It is therefore evaluated as a composition policy, not as a separately trained network.

4.2. Local Composition

The old plan, the predicted post-edit completion, and the edit support are composed through a control-change gate. The gate admits the requested delta only inside the declared region, then restores outside layers exactly. This operation is intentionally conservative: it can preserve an unsupported local change when the semantic execution is weak, but it cannot hide an outside violation behind a small average error.

4.3. Verification

FP-Eval checks mandatory validity and conditional watertightness. FEVP checks the five-layer outside contract and protocol attacks. Structural diagnostics report wall continuity, opening alignment, furniture containment, outside-footprint leakage, room overlap, and graph readouts. Raw watertightness and functional or componentwise watertightness are always reported separately.

5. Experimental Design

5.1. Populations and Exposure

Oracle-control results use CubiCasa5K test503 with 2,479 eligible edits across five operations and five repeated execution seeds. The primary local-delta path is deterministic on fixed rows; repeated seeds verify reproducibility rather than create independent floorplans.

Predicted structural reasonableness uses val491 with 491 target-free cases per seed and five repeated seeds. The operation audit contains 738 prompts from a development-exposed test manifest. Its row metadata declares `split=test`; legacy summary fields claiming val191 are ignored. A separate semantic diagnostic uses an older val191 model and is never pooled with the selected release model or with oracle rows.

Fresh held-out generalization is marked `optional_generalization_not_evaluated`. A new held-out claim would require a separate pre-access freeze, one execution, and retention of every negative result. It is not a prerequisite for the scoped claims in this paper.

5.2. Statistical Units

Delta-IoU intervals resample floorplans with replacement and average all edit instances inside a sampled floorplan.

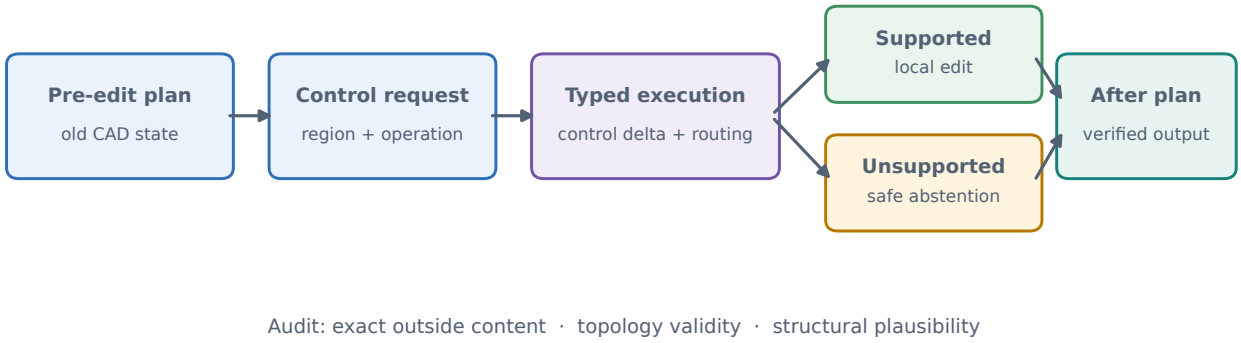


Figure 1. Strict local editing as a typed control transition. Supported requests are executed and audited; unsupported or ambiguous requests return a safe abstention. The outside contract is checked independently of edit fidelity.

Repeated seed rows are not treated as independent plans. Paired baselines use one difference per unique floorplan and Holm adjustment across five operations. The seed19 trace audit recomputes all 2,479 oracle rows with zero IoU error; other seeds remain reported reproducibility checks.

5.3. Matched Controls

The reconstruction wrapper applies Raster2Seq inside an outside-frozen local redraw. The editing-aware control consumes the same declared pre/post controls but recomputes its own paired FloorFlow row. Deterministic composition arms remove typed routing, dense cross-attention, edit gating, or the structural graph while keeping the measured rows fixed. The budget audit is post hoc and does not support an efficiency claim.

6. Results

6.1. Oracle-Control Editing Anchor

Table 1 reports the scoped oracle anchor. Room type, room addition, and room removal exceed 0.999 Delta-IoU, while opening deletion and furniture deletion reach 0.956703 and 0.987812. Outside exactness and FP-Eval validity are 1.0 for every operation. Pooled conditional watertightness is 0.957875–0.967491, and independent FEVP five-check validity ranges from 0.9124 to 0.9569.

These values measure binary change support. They do not prove requested-class correctness: a visually large change can have high Delta-IoU while realizing the wrong class. That boundary is why the paper reports semantic diagnostics separately rather than interpreting the oracle metric as deployable accuracy.

6.2. Target-Free Predicted Reliability

Table 3 separates structural plausibility from safety and semantics. Across val491, raster plausibility is 0.984929, plan plausibility 0.986558, FP-Eval validity 1.0, functional watertightness 0.997963, and wall continuity 0.995427. Room-face retention is 0.995655. Outside-footprint leakage and room overlap are zero.

Raw watertightness is 0.816701, not 1.0. Of 491 footprints, 401 are single-component and pass both raw and componentwise checks; 90 are multi-component and exceed the standard single-region FP-Eval boundary. The component-aware diagnostic does not redefine the raw metric. Predicted Room macro IoU is only 0.269701, confirming that structural reasonableness is not reference fidelity.

All 438 supported development-exposed prompts execute, all 300 unsupported or ambiguous prompts abstain, and every row preserves outside pixel and room-label content exactly with validity 1.0. This is a safety result. The older val191 semantic diagnostic remains mixed: room addition 92/100, room type 55/100, room removal 20/100, opening deletion 30/59, and furniture deletion 0/15. Object-delete and ghost-state contracts are separate execution checks and are not semantic successes.

6.3. Matched Controls and Composition

The outside-frozen Raster2Seq wrapper reaches Delta-IoU 0.137160–0.189491 across the five operations, while the editing-aware control reaches 0.731151–0.998508. FloorFlow's paired gains over the redraw wrapper range from 0.767212 to 0.862039. Gains over the editing-aware control range from 0.001340 for room removal to 0.245343 for furniture deletion. The small room-addition and room-removal gains are retained rather than rounded into a uniform improvement.

Typed routing leads the deterministic composition arms in Delta-IoU, Node F1, Edge F1, and GED. At matched observable routing budget it adds 0.048656 Delta-IoU and changes GED by -0.061396 with zero added learned parameters. It is slower: 3.1925 ms versus 0.2666 ms mean routing time. All 26 deterministic attacks are sensitive with observed

Table 1. Oracle-control editing anchor on CubiCasa5K test503. Intervals resample floorplans; repeated seeds are not independent plans. Delta metrics measure change support, not target-class correctness.

Operation	N	ΔIoU [95% CI]	ΔMSE	Outside exact	Vector validity
Room type	487	0.999199 [0.998420, 0.999747]	0.000069	1.000000	0.956879
Room add	495	0.999801 [0.999545, 0.999990]	0.000030	1.000000	0.955556
Room remove	500	0.999756 [0.999479, 0.999966]	0.000019	1.000000	0.954000
Opening delete	502	0.956703 [0.941316, 0.970424]	0.000096	1.000000	0.912351
Furniture delete	495	0.987812 [0.982565, 0.992375]	0.000083	1.000000	0.951515

Pooled FP-Eval validity 1.000; conditional watertightness 0.9579–0.9675. FEVP five-check validity 0.9124–0.9569.

Table 2. Same-case matched controls and deterministic composition-policy ablation. Raster2Seq is an outside-frozen wrapper and the editing-aware control is intentionally weak. The budget audit is post hoc and does not claim efficiency.

Operation	N	FloorFlow	Raster2Seq	Editing-aware	Gain vs R2S	Gain vs editing
Room type	487	0.999199	0.137160	0.909800	0.862039	0.089486
Room add	495	0.999801	0.166756	0.995985	0.833044	0.004015
Room remove	500	0.999756	0.177266	0.998508	0.822490	0.001340
Opening delete	502	0.956703	0.189491	0.731151	0.767212	0.225552
Furniture delete	495	0.987812	0.145367	0.742469	0.842445	0.245343

Arm	Outside px	Outside room	Violation	ΔIoU	Node F1	Edge F1	GED
Typed routing	1.000000	1.000000	0.000000	0.937473	0.955068	0.890007	2.120855
Dense routing	0.999545	0.999470	0.000455	0.887741	0.953797	0.887080	2.185478
No edit gating	0.986115	0.983599	0.013885	0.853715	0.949057	0.874121	2.448649
No structural graph	1.000000	1.000000	0.000000	0.888817	0.953819	0.887200	2.182251

Typed routing: ΔIoU gain +0.048656 and GED change -0.061396 at matched routing budget, with 3.1925 vs 0.2666 ms mean routing latency. Efficiency is not claimed.

false-pass and false-fail rates of zero.

6.4. External Context

Table 4 is intentionally not a leaderboard. The published HouseMind-E Delta-IoU and Delta-MSE anchors are 0.608 and 0.0019, numerically below the oracle-control pair, but the datasets, inputs, and evaluation units differ. HouseMind-E graph metrics remain substantially stronger than the current typed diagnostic. Predicted node F1 is 0.660202, edge overlap 0.264622, and GED 11.600815. We report this as a fidelity limitation, not a hidden zero.

7. Discussion

The results support a distinction between edit reliability and reference fidelity. Oracle controls show that the composition and restoration path can support large local changes while leaving the outside domain exact. Predicted controls show that the same path preserves the safety contract and produces geometrically plausible plans even when room identity and graph fidelity are weak. This is useful for an interactive system because unsafe edits and unsupported requests can be separated from semantic quality rather than averaged into one number.

The evidence also exposes the cost of that conservatism. Semantic realization is uneven, predicted Room macro IoU is low, and the raw watertightness boundary is visible for multi-component footprints. A system that appears reliable by preserving the input too aggressively is not sufficient if it fails to realize the requested operation. We therefore report execution and abstention as safety outcomes, not as semantic success.

8. Limitations

Oracle controls are target-derived and non-deployable. Predicted results come from an existing validation and development-exposed test exposure, not a fresh held-out run. Five repeated seeds are not independent plans. Raster2Seq is a reconstruction wrapper, not a native editing baseline. The editing-aware control is intentionally weak. Raw and functional watertightness answer different questions. HouseMind-E is cross-dataset context and the official serialization is unavailable.

The paper does not claim complete-method SOTA, RPLAN or full-CAD leadership, deployable free-text control, high predicted room identity, or uniform semantic success. Move-wall, resize, and free-form global recomposition

Table 3. Target-free predicted-control evidence. (A) Structural reasonableness on val491 across five repeated seeds. (B) Development-exposed safety prompts and a separately labeled older-model semantic diagnostic. Raw and functional watertightness are distinct.

Metric	Mean
Raster plausible	0.984929
Plan plausible	0.986558
Room coverage	1.000000
Room-face retention	0.995655
FP-Eval validity	1.000000
Raw watertight	0.816701
Functional watertight	0.997963
Wall continuity	0.995427
Opening-wall align	0.993530
Furniture containment	1.000000
Outside-footprint fraction	0.000000
Room overlap ratio	0.000000
Room macro IoU (diag.)	0.269701

Safety evidence	Observed	
Supported executed	438/438	
Unsupported abstained	300/300	
Outside pixel exact	1.000000	
Outside room exact	1.000000	
Validity	1.000000	
Semantic diagnostic	**Success / N**	**Rate**
Room add	92/100	0.9200
Room type	55/100	0.5500
Room remove	20/100	0.2000
Opening delete	30/59	0.5085
Furniture delete	0/15	0.0000

Semantic rows use an older val191 model and are not a release gate. The safety population is development-exposed test, not held out.

Table 4. External context and fidelity boundary. Rows use different datasets, inputs, and protocols and must not be read as a leaderboard.

Source	Task / input	ΔIoU	ΔMSE	Node F1	Edge	GED	Interpretation
FloorFlow oracle	CubiCasa test503 oracle controls	0.956703–0.999801	0.000019–0.000096	–	–	–	Scoped anchor
FloorFlow typed	12,395 strict-locality rows	0.937473	–	0.955068	0.890007	2.120855	Diagnostic
FloorFlow predicted	val491 target-free controls	–	–	0.660202	0.264622	11.600815	Weak fidelity
HouseMind-E	RPLAN editing, published	0.608000	0.001900	0.998000	0.934000	0.467000	External context
Raster2Seq wrapper	CubiCasa frozen redraw	0.137160–0.189491	–	–	–	–	Reconstruction wrapper

FloorFlow oracle ΔIoU/ΔMSE numerically exceed the published HouseMind-E pair under different datasets and inputs. Graph and predicted-fidelity metrics are not uniformly superior; predicted room macro IoU is 0.269701.

remain outside the supported operation set.

Data and Ethics

The study uses CubiCasa5K under its research-use terms. It introduces no human subjects, participant data, or personally identifiable information. The release excludes datasets, checkpoints, raw tensors, third-party source trees, and unselected generated outputs. The evidence registry, frozen file identities, figure case manifest, artifact manifest, and checksums are released with the manuscript.

Reproducibility Statement

Every numerical value in the main tables is bound to `docs/floorplan_submission_source_registry_v2.json` and the frozen release in `configs/floorplan_submission_evidence_freeze_v2.json`. The provenance and freeze audits are machine-readable. Figure cases are selected by a recorded deterministic rule and stored in `outputs/reports/floorplan_figure_case_manifest_v1.json`. The artifact package is anonymous and contains source, tests, selected configurations, manuscript sources, tables, figures, and selected evidence. It does not contain datasets or checkpoints.

LLM Usage Statement

Large language model assistance was used for code editing, artifact auditing, LaTeX formatting, and language editing. Numerical claims were checked against the frozen source registry and generated tables. The reported experiments were not replaced by model-generated results, and no held-out number, execution receipt, or venue submission receipt was fabricated. The authors remain responsible for the final manuscript, data use, and external submission.

References

[1] Cubicasa5k: A dataset and benchmark for floorplan recognition. In *European Conference on Computer Vision Workshops*, 2019. 2

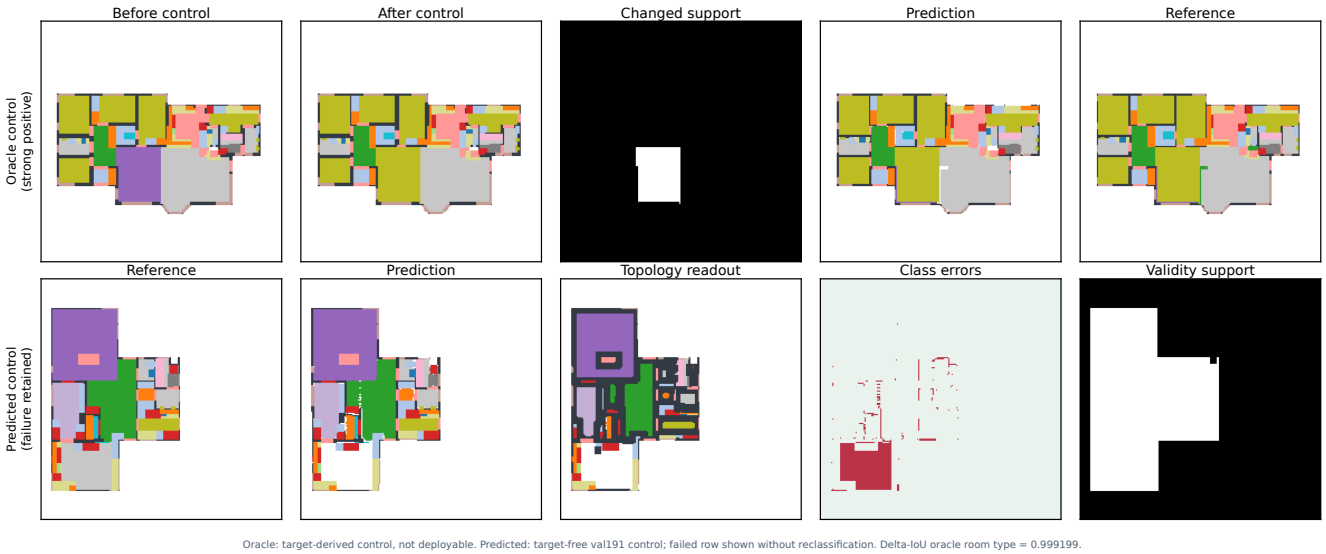


Figure 2. Case-level audit. Top: oracle room-type control with before/after controls, changed support, prediction, and reference. Bottom: a target-free predicted failure retained without reclassification. Oracle controls are target-derived; the predicted case is not held out.

[2] Gsdiff: Geometry-enhanced structural graph generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025. 2

[3] Ce2eplan: End-to-end boundary-conditioned vector floorplan generation. *IEEE Transactions on Visualization and Computer Graphics*, 2026. 2

[4] Housemind: Unified floorplan understanding, generation, and editing, 2026. Public protocol anchors used only as cross-dataset context. 1

[5] Raster2seq: Vector floorplan reconstruction from raster inputs, 2026. Matched reconstruction wrapper in this study. 2